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- 11 D** Let v be the velocity of the ball at the bottom of the circle

Gain in ke = loss in gpe

$$\frac{1}{2} m v^2 = mg (L - L \cos 60^\circ)$$

$$v^2 = Lg$$

At the bottom:

$$T - mg = mv^2 / L = m(Lg)/L = mg$$

$$T = 2 mg$$

- 12 D** Since net gravitational force is zero